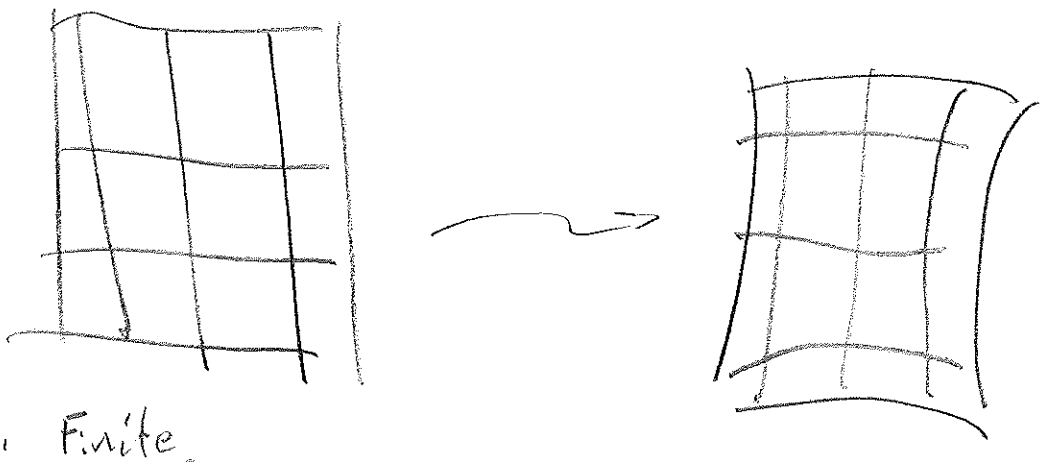
① Linear Solvers - 3D

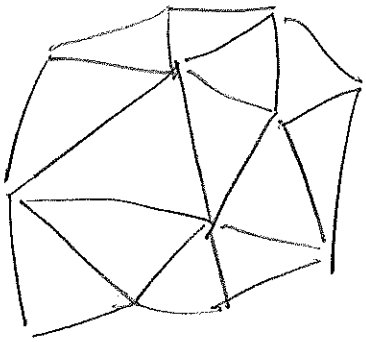
FEM leads to unstructured graph

Structured



- Finite Diff
- Finite Volume
- Spectral

$A = B_y \otimes A_x + A_y \otimes B_x$



→ FEM - unstructured

A - unstructured sparse matrix

(2)

For FEM - solver options are

- direct: $\underline{u} = A^{-1}\underline{b} = A \backslash \underline{b}$

- Iterative

$$\underline{u}_0 = \underline{0}$$

$$\underline{u}_{k+1} = \underline{u}_k + M^{-1}(\underline{b} - A\underline{u}_k) \quad (\text{say})$$

Requires only $\underline{w} = A\underline{p}$ } sparse matvec (SpMV)

and solve

$$M\underline{z} = \underline{r}$$

$M :=$ preconditioner { $\approx A$ and easy to solve

Assume cost of

$$M\underline{z} = \underline{r} \rightarrow \underline{z}$$

is $\approx$ same as cost of $\underline{w} = A\underline{p}$

③

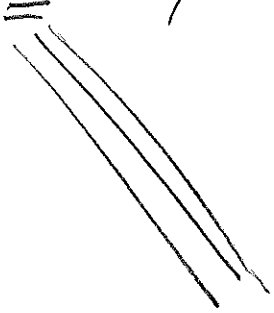
- Usually, $\underline{w} = A_f$ is leading-order cost and scales as #nonzeros in A

- Consider finite diff., n unknowns

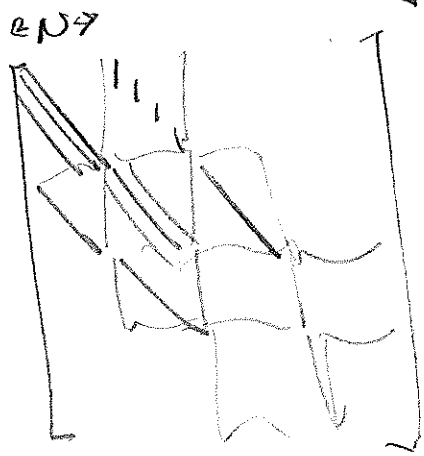
			FEM
1-D	nnz	$\simeq 3 \cdot n$	$3n$
2-D	"	$\simeq 5n$	$9n$
3-D	"	$\simeq 7n$	$27n$

- Matrix Bandwidth β, $N \times N \times N$ grid

1-D $\beta = 1$



2-D



$\beta = N$

3D · $\beta = N^2$

For banded systems, direct solver cost is $ops \simeq n \cdot \beta^2$

	1D	2D	3D
Banded	$8n$	$2n^2$	$2nN^4 = 2N^7 = 2n^2 n^{\frac{1}{3}}$
Optimal	$8n$	$4n^{3/2}$	$6n^2$
(nested dissection)	$8N$	$4N^3$	$6N^6$

- Resolution: $h \sim \frac{L}{N}$ $\quad n = N^d$

- For a given Resolution, h/L, costs are <u>much</u> higher for 3D than 2D

WHY:

	2D	3D
n	$= N^2$	N^3
Factor	$O(N^3)$	$O(N^6)$

$\leftarrow$ Because of Increased matrix Bandwidth!

(5)

- Iteration methods do not know or care about matrix bandwidth -

- Just $nnz(A)$

and $cond(A)$ or $\underline{\underline{cond(M^{-1}A)}} = K(M^{-1}A)$

- Cost:

cost-per-iteration $\times$ # iterations to meet tolerance

$(nnz(A))$ $\qquad$ $(\sim\sqrt{K(M^{-1}A)})$

- You generally NEED to be in $\mathbb{R}^3$ to see a clear win for iterative methods.

(Table from two_level)

(6)

- Suppose A & M are SPD —

Then, best iterative method is

Preconditioned Conjugate Gradients (PCG)

$A\underline{x}=\underline{b}$; $\underline{r}=\underline{b}$, $\underline{p}=\underline{0}$, $\rho_1=1$

for $k=1,\ldots$

$\underline{z}=M^{-1}\underline{b}$ $\quad \rho_0=\rho_1 \quad \rho_1=\underline{r}^T\underline{z} \quad \beta=\rho_1/\rho_0$

$\underline{p}=\underline{z}-\beta\underline{p}$

$\underline{w}=A\underline{p} \quad \gamma=\underline{w}^T\underline{p} \quad \alpha=\rho_1/\gamma$

$\underline{x}=\underline{x}+\alpha\underline{p}$

$\underline{r}=\underline{r}-\alpha\underline{w} \qquad (\equiv \underline{b}-A\underline{x}\equiv A\underline{e})$

stop when $\|\underline{r}\|_\infty \le tol\,\|\underline{b}\|$ (say)

end

- Four vectors
  - 1 matvec
  - 1 preconditioner solve.

(2)

- Consider Jacobi iteration $A\underline{x} = \underline{b}$

$$\underline{x}_0 = 0$$

$$\underline{x}_{k+1} = \underline{x}_k + D^{-1}(\underline{b} - A x_k) \qquad D \equiv M$$

- Let's assume that $D = M = I$

$$\underline{s}_k = \underline{b} - A\underline{x}_k = A\underline{e}_k \qquad e_k = \underline{x} - \underline{x}_k$$

$$\underline{x}_{k+1} = \underline{x}_k + \underline{s}_k$$

$$\underline{x}_0 = 0$$

$$\underline{x}_1 = \underline{b}$$

$$\underline{x}_2 = \underline{b} + (\underline{b} - A\underline{b}) = (2 - A)\underline{b}$$

$$\underline{x}_3 = (2 - A)\underline{b} + (\underline{b} - A(2-A)\underline{b})$$

$$= 3\underline{b} - 3A\underline{b} + A^2\underline{b} \qquad \in \mathbb{P}_2(A)\underline{b}$$

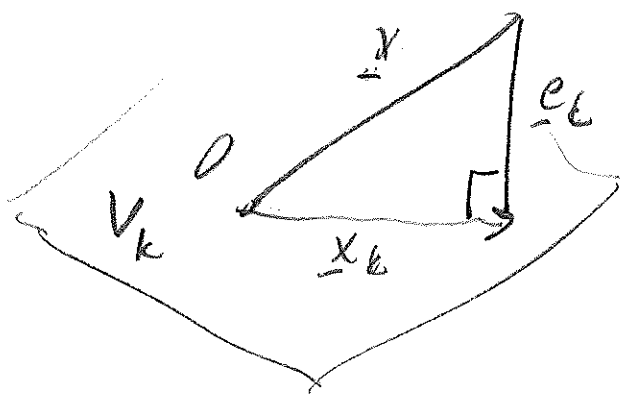
$$K_k(A, b) = \text{span}\{\underline{b}, A\underline{b}, \ldots, A^{k-1}\underline{b}\}$$

$$\underline{x}_k \in K_k(A, \underline{b})$$

(8)

- Projection methods find best fit in K_k

- Consider L^2 projection of $\underline{x}$ onto V_k



$$\underline{x}_k = \sum_{j=1}^{k} \beta_j v_j$$

Least squares $\quad \underline{e}_k \perp \underline{v}_j \quad j=1,\dots,k$

$$V^T(\underline{x} - \underline{x}_k) = 0$$

$$V^T \underline{x}_k = V^T \underline{x}$$

Formally, use normal eq'ns.

(9)

$$\underline{x}_k = V\underline{\beta}$$

$$V^T V \underline{\beta} = V^T \underline{x}$$

- $\underline{x}_k = V(V^T V)^{-1} V^T \underline{x}$
- Better if $(\underline{v}_i, \underline{v}_j) = \delta_{ij}$
- Easy, by Gram-Schmidt $i = 1, \dots, k$

$$\underline{v}_i = \underline{v}_i - V_{i-1} V_{i-1}^T \underline{v}_i$$

$$\underline{v}_i = \underline{v}_i / \sqrt{(\underline{v}_i, \underline{v}_i)}$$

(10)

Iteration — given $(\underline{r}_0 \cdots \underline{r}_{k-1})$.

$V_0 = 0 \qquad \underline{x}_0 = 0$

for $j = 1, \ldots, k$

$$\underline{v}_j = \underline{r}_{j-1}$$

$$\underline{v}_j' = \underline{v}_j - V_j V_j^T \underline{v}_j \qquad \underline{v}_j = v_j' / (\underline{v}_j, \underline{v}_j)^{\frac{1}{2}}$$

$$\underline{x}_j = \underline{x}_{j-1} + \frac{(\underline{v}_j, \underline{x})}{\underbrace{(\underline{v}_j, \underline{v}_j)}_{1}} \underline{v}_j$$

end

- At each k, $\underline{x}_j$ is closest element to $\underline{x}$ in $R(\underline{w}_1 \cdots \underline{w}_j)$

- Update does <u>not</u> require refactor of $j \times j$ matrix $V_j^T V_j$

(11)

- In our case, don't know $\underline{x}$! (2)

- But, we do not need it!

$\Rightarrow$ Change inner products to be computable

$$\frac{(v_j, \underline{x})}{(v_j, v_j)} \rightarrow \frac{(v_j, \underline{x})_A}{(v_j, v_j)_A} = \frac{\underline{v}_j^T A \underline{x}}{\underline{v}_j^T A \underline{v}_j} = \frac{\underline{v}_j^T \underline{b}}{\underline{v}_j^T A \underline{v}_j}$$

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for $k = 1, \ldots, k$

$\underline{v}_k = \underline{r}_{k-1}$

Short term recurrence!
$\beta_j = 0 \quad j = 1, \ldots, k-2$

$\underline{v}_k = \underline{v}_k - \sum_{j=1}^{k-1} \underline{v}_j \beta_j \qquad \beta_j := \underline{v}_j^T A \underline{v}_{k-1}$

$\underline{x}_k = \underline{x}_{k-1} + \alpha_k \underline{v}_k \qquad \alpha_k = \frac{\underline{v}_k^T \underline{b}}{\underline{v}_k^T \underline{w}_k} \qquad \underline{w}_k := A \underline{v}_k$

$\underline{r}_k = \underline{r}_{k-1} - \alpha_k \underline{w}_k$

end

Summary: CG –

(1) Short term recurrence –

$$\underline{v}_k = \underline{p}_k \in K_k(A, \underline{b})$$

<u>FIXED-COST</u> per iteration (Yay!)

(2) $\underline{x}_k$ is <u>closest</u> element to $\underline{x}$ in $K_k(A, \underline{b})$, in the A-norm:

$$\| \underline{v} \|_A = (\underline{v}^T A \underline{v})^{\frac{1}{2}} \qquad \underline{A\text{-SPD}}$$

(3) ADVANTAGE manifest <u>only</u> for $d \gtrsim 2$!

(13)

Convergence bound:

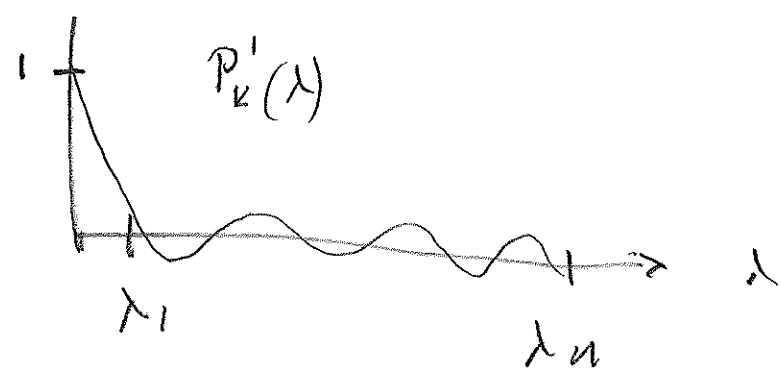
$$\|\underline{e}_k\|_A^2 := \|\underline{x} - \underline{x}_k\|_A^2 \leq \|\underline{x} - \underline{v}\|_A^2 \quad \forall\, \underline{v} \in K_k(A, \underline{b})$$

- Sketch –

$$\underline{r}_k := \underline{b} - A\underline{x}_k \equiv A\underline{e}_k$$

$$\underline{r}_k = \underline{b} + a_1 A\underline{b} + \cdots A^k \underline{b} \in \mathbb{P}_k^1(A)\cdot\underline{b}$$

$$\mathbb{P}_k^1(\lambda) = \{\mu \text{ st } \mu \in \mathbb{P}_k(\lambda)\ \mu(\lambda = 0) = 1\}$$



(14) NOTE: $A\underline{x} = \underline{b} \iff x = A^{-1}\underline{b}$ $\quad \|\underline{x}\|_A = \|\underline{b}\|_{A^{-1}}$

$$\|\underline{e}_k\|_A^2 = \|\underline{r}_k\|_{A^{-1}}^2 \le \|P_k'(A)\,\underline{b}\|_{A^{-1}}^2 \quad \forall\, P_k'(x) \in \mathbb{P}_k'(x)$$

Suppose $A\underline{z}_j = \lambda_j \underline{z}_j$, $\underline{z}_i^T \underline{z}_j = \delta_{ij}$

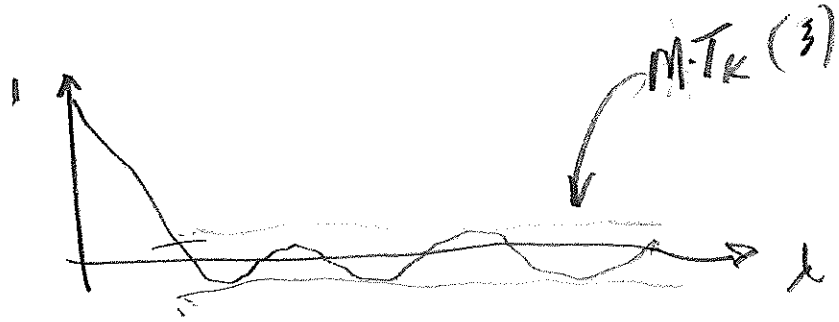
$$\underline{b} = \sum_{j=1}^{n} \hat{b}_j \underline{z}_j$$

$$(P_k'(A)\underline{b})^T A^{-1} (P_k'(A)\underline{b})$$

$$= \sum_{j=1}^{n} \frac{1}{\lambda_j} P_k'(\lambda_j)^2 \hat{b}_j^2 \le M^2 \|\underline{x}\|_A^2$$

$$M := \max_j |P_k'(\lambda_j)| \le \max_{\lambda \in (\lambda_1, \lambda_n)} |P_k'(\lambda)|$$

For <u>all</u> $P_k'(\lambda)$!



(15)

$$M \simeq 2\left(\frac{\sqrt{K}-1}{\sqrt{K}+1}\right)^k$$

$$\therefore \|\underline{e}_k\|_A \leq 2\left(\frac{\sqrt{K}-1}{\sqrt{K}+1}\right)^k \|\underline{x}\|_A$$

- Preconditioning
  - Go-to preconditioner (when in a hurry)

$$M^{-1}\underline{r} = D^{-1}\underline{r} + J A_c^{-1} J^T \underline{r}$$

$$A_c = J^T A J$$

- J = coarse-to-fine interpolant.

(curr_paper.pdf, eqn (3))

(17) Generally for FEM,

- $\lambda(D^{-1}A) \in [0,2]$ – Gershgorin + diagonal dominance.

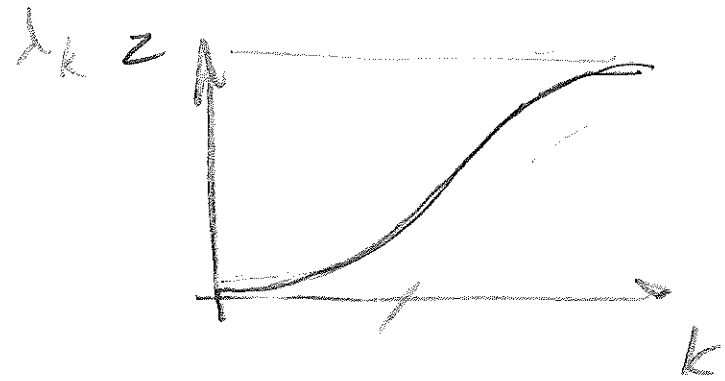
- Suppose

$$J = [\underline{\xi}_1 \cdots \underline{\xi}_m] \qquad A\underline{\xi}_i = \lambda_i \underline{\xi}_i \qquad m \ll n \qquad \lambda_1 \le \lambda_2 \le \lambda_n$$

- What is spectrum of $M^{-1}A$?
  - Take $A = \text{tridiag}(-1, 2, -1)$
  - Consider $M^{-1}A\underline{\xi}_k \quad k > m$

$$D^{-1}A\underline{\xi}_k = \lambda_k(D^{-1}A)\underline{\xi}_k$$

$$\lambda_k = 1 - \cos\theta_k \qquad \theta_k = \frac{\pi k}{N} \qquad k = 1, \ldots, N-1$$



(18)

For $k < m$

$$M^{-1} A \underline{z}_k = \Big[\underbrace{(1 - \cos\theta_k)}_{\lambda_k} + \frac{1}{\lambda_k} \underline{z}_k^T \underline{z}_k \lambda_k \Big] \cdot \underline{z}_k$$

$$= \lambda_k + 1$$

Check:

$$J = [\underline{z}_1 \cdots \underline{z}_m]$$

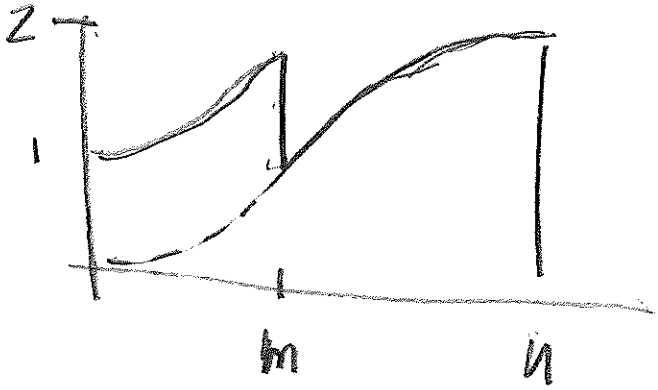
$$A_c = J^T A J = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_m \end{bmatrix}$$

$$J^T (A \underline{z}_k) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \lambda_k$$

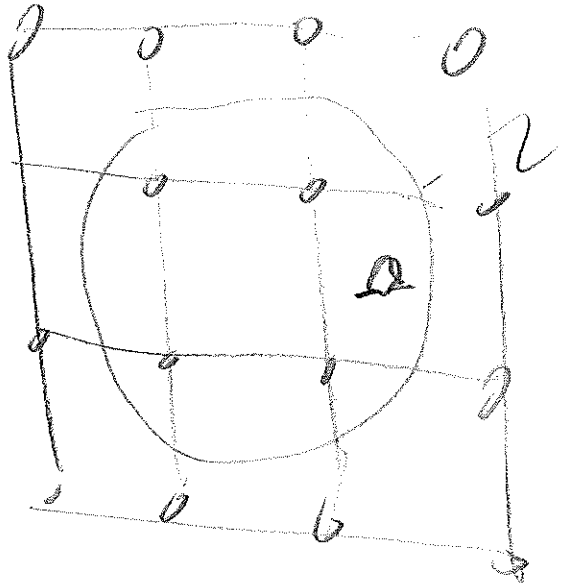
$$J A_c^{-1} J^T = \underline{z}_k$$

(19)

Spectrum is modified



$$\kappa(M^{-1}A) \simeq \frac{\lambda_n}{\lambda_m} \ll \frac{\lambda_n}{\lambda_1}$$



J = bilinear interpolants onto $x_j \in \bar{\Omega} \setminus \partial\Omega$

$\equiv \Omega$